

● EASY

● ADVANCED MATH

Advanced Math

04

10 questions · ~15 min

Q1

ADVANCED MATH

Which expression is equivalent to $15w^2 + 8w$?

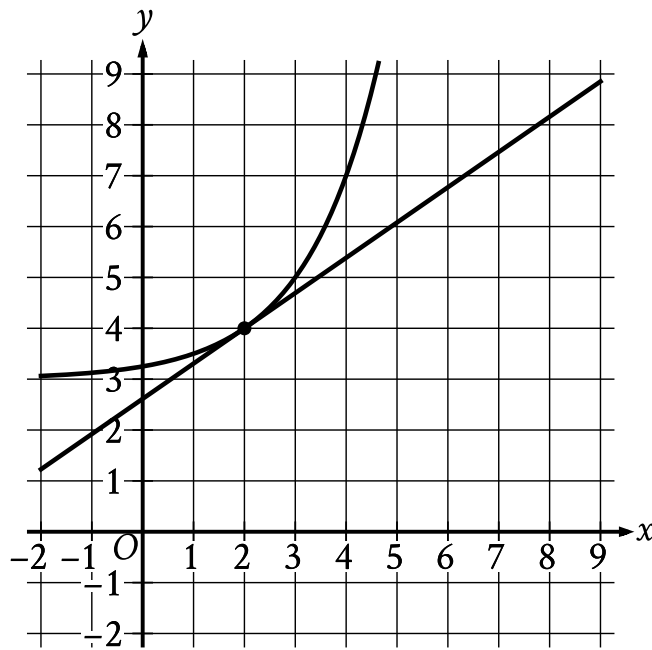
- A. $w15w + 8$
- B. $8w15w + 1$
- C. $15w^28w + 1$
- D. $23w^2 + w$

Q2

ADVANCED MATH

The function f is defined by $fx = \frac{16}{x}$. What is the value of fx when $x = 17$?

- A. $\frac{16}{17}$
- B. $\frac{17}{16}$
- C. 16
- D. 17



- For the curve in the system:
 - Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point (0 comma 3.25).
 - In quadrant 1, the curve trends up sharply.
 - The curve passes through the following points:
 - (1 comma 3.5)
 - (2 comma 4)
 - (4 comma 7)
- For the line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (2 comma 4)
 - (7 comma 7.5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

- A. 0,0
- B. 0,2
- C. 2,4
- D. 4,0

Q4

ADVANCED MATH

Which of the following expressions is equivalent to $2(ab - 3) + 2$?

- A. $2ab - 1$
- B. $2ab - 4$
- C. $2ab - 5$
- D. $2ab - 8$

The function $f(x) = 200,000(1.21)^x$ gives a company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online, where $0 < x \leq 10$. What is the best interpretation of the statement " $f(5)$ is approximately equal to 518,748" in this context?

- A. 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately 518,748 dollars.
- B. 5 years after the company started selling light bulbs online, its predicted annual revenue will have increased by a total of approximately 518,748 dollars.
- C. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5% greater than the predicted annual revenue for the previous year.

$$3x^2 - 15x + 18 = 0$$

How many distinct real solutions are there to the given equation?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Q7

ADVANCED MATH

Which expression is equivalent to $6x + 5x + 4y$?

- A. $15x$
- B. $15y$
- C. $11x + 4y$
- D. $30x + 4y$

Q8

ADVANCED MATH

If $f(x) = \frac{x^2 - 6x + 3}{x - 1}$, what is $f(-1)$?

- A. -5
- B. -2
- C. 2
- D. 5

Q9

GRID-IN

ADVANCED MATH

$$x^2 = 2222$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which expression is equivalent to $20w - 4w + 3w$?

A. $10w$

B. $13w$

C. $19w$

D. $21w$